

Anish Anandhan A L

OT/ICS Security | Cyber Defense | Cloud Security

Hyderabad, India | +91 8870676412 | anishanandhan13@email.com | [LinkedIn](#) | [GitHub](#) | [Substack](#) | [Medium](#)

SUMMARY

Specializing in OT/ICS Security, Cyber Defense, and Vulnerability Management, with hands-on experience across vulnerability assessment, security operations, network security, and threat intelligence. Skilled in OT/ICS security architecture, IEC 62443, industrial protocols, OT vulnerability assessment and penetration testing, CVE/CVSS analysis, SIEM, and security automation. Building deeper expertise in Cloud Security, security research, reverse engineering, and vulnerability analysis.

EXPERIENCE

OT/ICS Security Intern

Jul 2026 – Present

Medtronic PLC

Hyderabad, India

- Contribute to **OT Security** initiatives across security governance, management, PMO, vulnerability management, and cyber defense functions.
- Support **OT vulnerability management** using Armis and Rapid7 for asset visibility, vulnerability identification, risk assessment, and remediation tracking across industrial environments.
- Contribute to **OT Cyber Defense** activities including incident response (IR) and cyber threat intelligence (CTI), supporting security monitoring and investigation workflows.
- Automated an **IOC-driven vulnerability assessment workflow** using Microsoft Power Automate, enabling IP addresses or hostnames received as IOCs to be automatically investigated for associated vulnerabilities.

Research Intern

May 2026 – July 2026

Centre for Cybersecurity, Trust and Reliability, IIT Madras

Chennai, India

- Contributed to **DriverArmory**, a research-grade BYOVD intelligence and exploitation platform for analyzing signed vulnerable Windows drivers.
- Worked on the **Command and Control (C2) component**, integrating Cipher C2 with BOF/COFF-based in-process payload dispatch and the DriverArmory backend.
- Developed a **Ghidra headless analysis pipeline** to automate driver binary analysis, IOCTL mapping, vulnerability pattern detection, and CVE cross-referencing.
- Worked with a corpus of **400+ signed vulnerable drivers**, extracting security-relevant metadata and generating vulnerability analysis reports.

TECHNICAL SKILLS

OT/ICS Security: IEC 62443, ICS/SCADA, PLC, HMI, Industrial Protocols, OT Vulnerability Assessment, OT Penetration Testing

Cyber Defense: Vulnerability Management, Incident Response, Threat Intelligence, IOC Analysis, SIEM, Security Monitoring

Security: CVE/CVSS Analysis, Network Security, Web Application Security, Security Testing

Security Tools: Armis, Rapid7, Qualys VMDR, Wireshark, Nmap, Burp Suite, OWASP ZAP, ELK/Kibana

Cloud & DevSecOps: AWS, Docker, Git, SAST/DAST, Security Automation

Security Research: Reverse Engineering, Ghidra, Vulnerable Driver Analysis, Firmware Security, Fuzzing

PROJECTS

OTSecTwin - OT/ICS Security Digital Twin | *Python, OpenPLC, Modbus/TCP, ELK, Docker* 2026

- Built an **OT/ICS security digital twin** for a simulated industrial environment, modeling PLC, HMI, SCADA assets, communication paths, and network topology.
- Implemented **OT asset discovery and vulnerability correlation**, mapping observed assets to CVE/CVSS data and incorporating asset criticality, exposure, exploitability, and process impact into risk assessment.
- Developed behavioral baselines and detection logic to identify **unauthorized PLC writes, anomalous Modbus function codes, new communication pairs, network scans, and unknown assets**.
- Built a security dashboard using **React and Elasticsearch/Kibana** to visualize OT assets, topology, vulnerabilities, security events, and risk insights.

- AetherCTI - Cyber Threat Intelligence & Attribution Platform | *SQLite, MITRE ATT&CK*

2026
- Built a **CTI platform** for automated analysis of IP addresses, file hashes, domains, and URLs by querying **80+ security engines and intelligence sources** in parallel.
 - Developed an asynchronous intelligence aggregation engine using **FastAPI, asyncio, and httpx** to correlate vendor verdicts, abuse reports, CVEs, geolocation, and threat intelligence into a unified attribution matrix.
 - Implemented a transparent **0–100 threat scoring engine** and automated **MITRE ATT&CK** TTP mapping to support SOC investigation and threat assessment workflows.

CERTIFICATIONS

Microsoft Certified: Security Operations Analyst Associate (SC-200)

Issued 2026

ISC2 Certified in Cybersecurity (CC)

Issued Jan 2026

AWS Certified CloudOps Engineer – Associate

Issued 2026

RESEARCH PUBLICATIONS

Integration of Piezoelectric and LDR Sensors for Efficient Electricity Generation

2025

IEEE Conference Publication

- Proposed an energy-efficient system integrating piezoelectric and LDR sensors for sustainable electricity.
- Demonstrated applications in smart infrastructure and energy optimization.
- **DOI:** 10.1109/ICTMIM65579.2025.10988268

Unsupervised Anomaly Detection using Deep Feature Extraction and Clustering

2025

IEEE Conference Publication

- Developed an unsupervised learning model for anomaly detection using deep feature extraction.
- Applied machine learning techniques to identify irregular patterns in large-scale datasets.
- **DOI:** 10.1109/ICCRTEE64519.2025.11053096

EDUCATION

Vellore Institute of Technology, Chennai, India

Mtech Integrated Software Engineering, School of Computer Science and Engineering

2022 - 2027